

Inflorescence

- **Inflorescence**

- Flower is a reduced shoot which is the reproductive organ of a plant, and it develops either from an axillary or from a terminal bud.
- A floral axis or peduncle can have multiple flowers borne on it. Flowers can be borne singly or in clusters. If borne singly, flowers are said to be solitary.
- Flowers borne in clusters are arranged in different patterns on the peduncle.
- The arrangement of flowers on the floral axis is known as **inflorescence**.

- Inflorescence can primarily be of five types:
- Solitary inflorescence
- **Racemose** inflorescence
- **Cymose** inflorescence
- Mixed inflorescence
- Specialized inflorescence

A. Solitary flowers

The flowers which occur singly or are separated from other flowers of the same plant by vegetative regions are called Solitary flowers. Solitary flowers are of two types:

- a. Solitary terminal: Single flower occur at the tip of the main stem and its branches. E.g. Poppy.
- b. Solitary axillary: Single flower occurs in the axil of foliage leaves e.g. shoe flower.

- **Racemose inflorescence**

- In racemose inflorescence, there is indefinite growth of the main axis as it does not terminate into a flower. The flowers arrangement is in **acropetal order**, i.e, older flowers lie at the base and younger flowers at the apex. Example: Gulmohar

In circular arrangement, centripetal succession, i.e. older towards the periphery and younger towards the center.

- Racemose inflorescence can be of certain types which are listed below:
- **Simple racemose inflorescence**
- Racemose inflorescence is said to be simple if the peduncle is unbranched. This can be of the following types -
- **Simple raceme**
- The main axis is unbranched, elongated and bears flowers laterally in an acropetal succession in simple raceme. There are many pedicellate flowers produced. Example: *Crotalaria*.

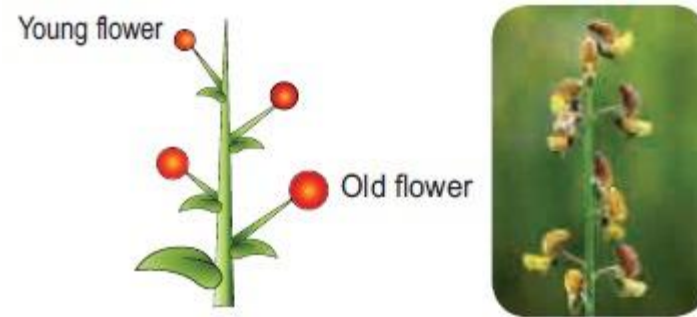
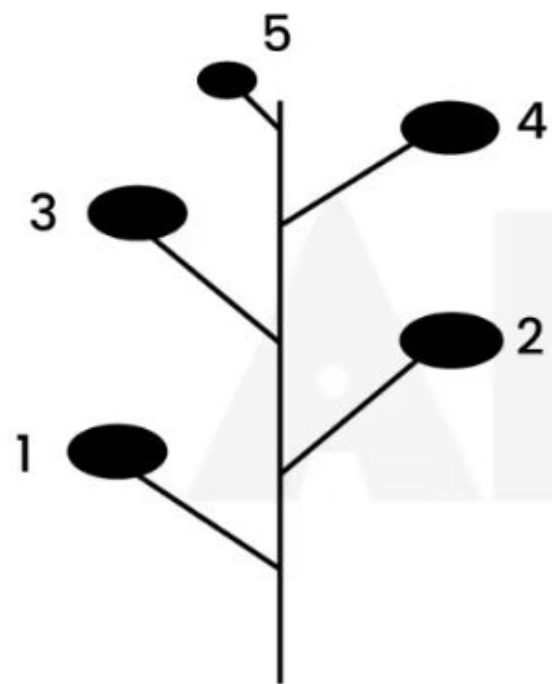
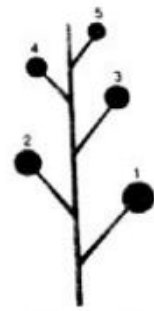


Figure 4.4: (a) diagrammatic,
(b) Simple raceme



Raceme





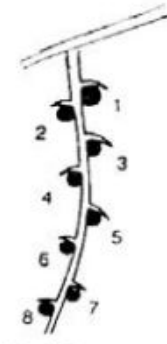
Raceme



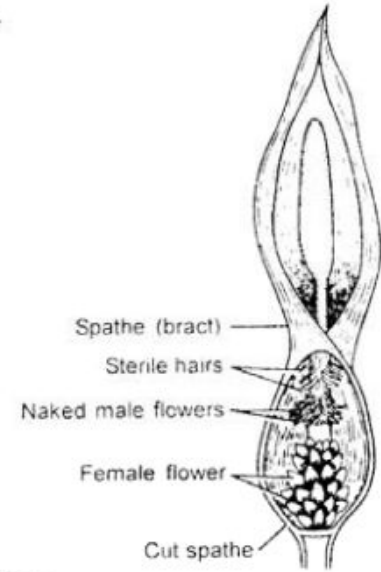
Panicle



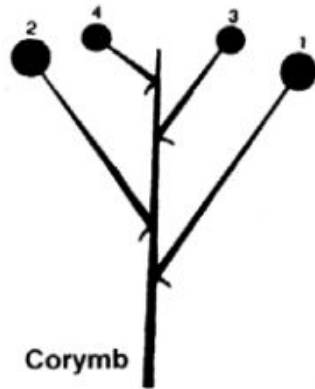
Spike



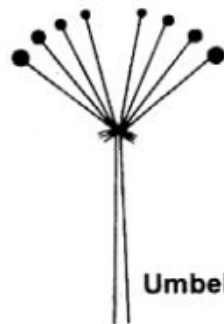
Catkin



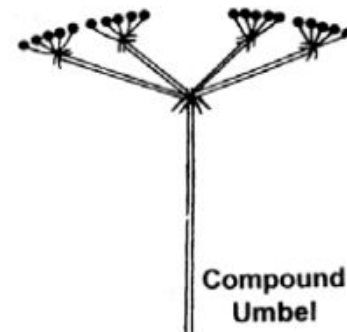
Spadix



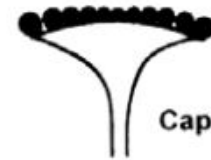
Corymb



Umbel



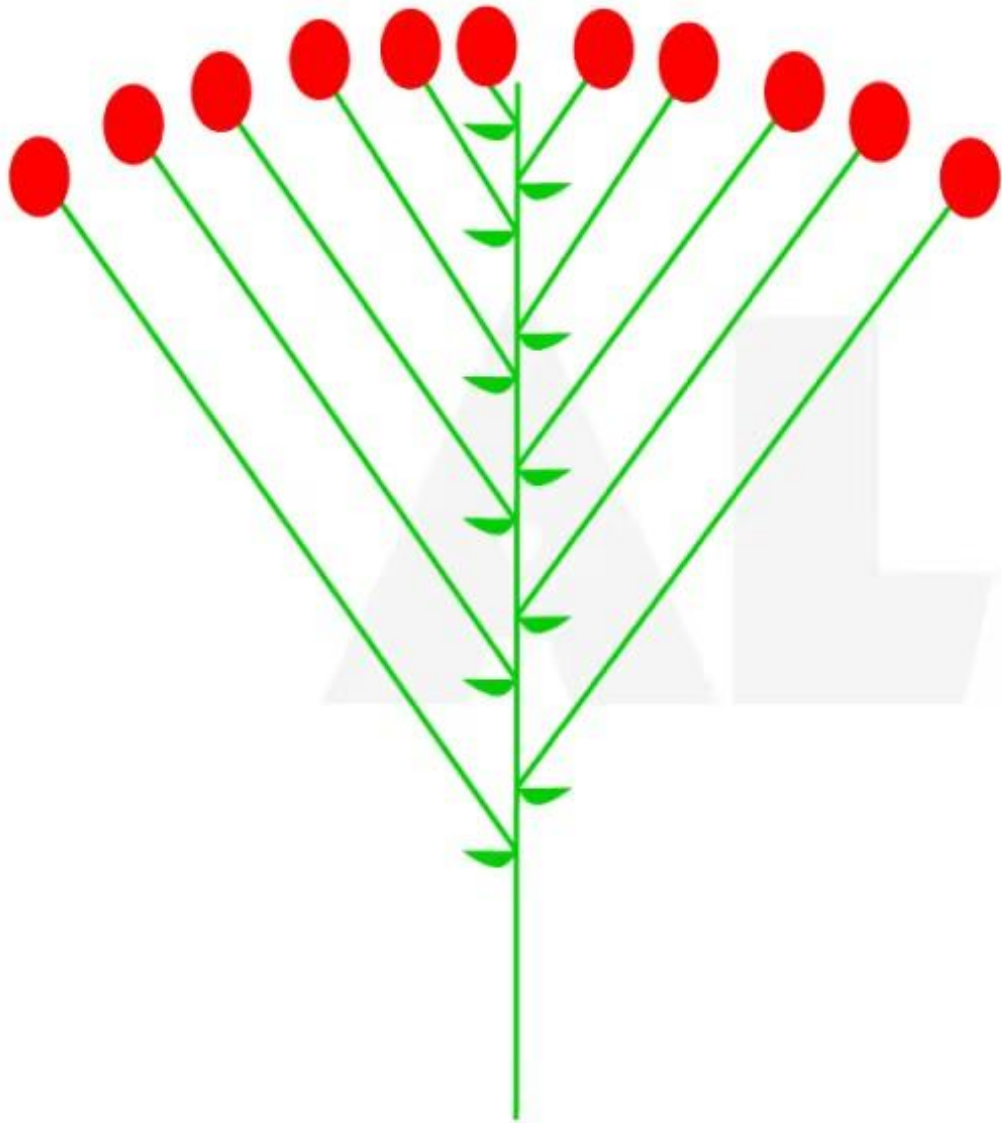
Compound Umbel



Capitulum

- **Corymb**

- In corymb, the peduncle is short and unbranched flowers are pedicellate. The lower flowers have longer stalked than the upper ones. All flowers lie at the same level.
Example: *Iberis* (candytuft).



- **Umbel**

- In this type of inflorescence, the **peduncle** is short and condensed. The flowers arise in clusters from the same point. The **pedicels** of all the flowers are the same length. A whorl of **bracts** known as involucre may be present at the base.
Example: Onion

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- **Spike**

- Main **axis** is **elongated** in a spike. **Sessile flowers** are arranged in **acropetal** fashion.

Examples: *Amaranthus*, *Callistemon* (bottlebrush) etc.

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- **Catkin**

- It is similar to a spike but has a **long pendulous axis**. Flowers borne on the main axis are **unisexual** and **sessile**. Example: Mulberry

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Figure 4.4: (g) diagrammatic, (h) Catkin

- **Spadix**

- It is a **spike** with a **long and fleshy peduncle**. The peduncle is covered with one or more colored bracts called **spathe**.
The **male** and **female flowers** are **closely packed** together.
The base has female flowers while the male flowers occupy the upper region of the peduncle. Sterile florets lie in the middle.
The spathe becomes tubular at the base. Examples: Maize, banana, *Colocasia*, etc.

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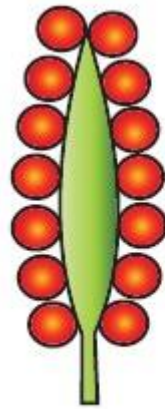


Figure 4.4: (i) diagrammatic, (j) Spadix

- **Compound racemose inflorescence**

- It is the type of racemose inflorescence in which the peduncle is branched, and the branches are borne on the mother peduncle in an acropetal fashion. It is of the following types -

- **Compound raceme**

- Pedicellate flowers are borne acropetally on the lateral branches of an elongated peduncle.



Figure 4.4: (k) diagrammatic, (l) Panicle

- **Compound spike**

- Sessile flowers are borne over branches of elongated peduncles.

- **Compound spadix**

- The unisexual sessile flowers are borne over the branches of the peduncle of a spadix. The whole inflorescence is covered by a spathe and the individual branches may or may not be covered by spathes known as spathellae.

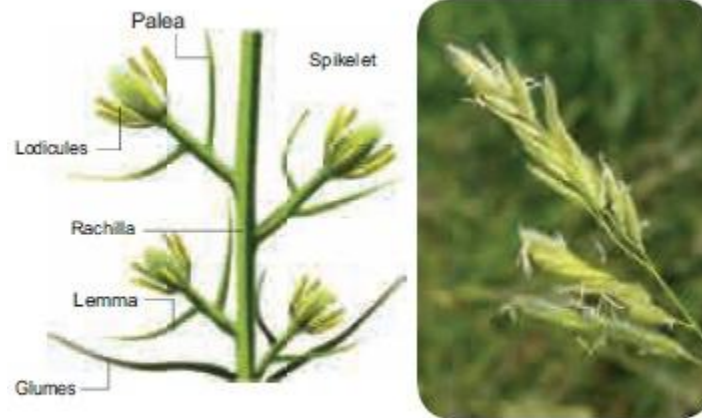


Figure 4.4: (e) diagrammatic, (f) Spikelet

- **Compound corymb**
- A mother corymb bears a number of daughter corymbs.

- **Compound umbel**

- A number of daughter umbels known as umbellules are formed from a single point which has an involucre of bracts at its base. Daughter involucres of bracts or involucels can be seen at the base of each umbellule.

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Figure 4.4: (m)
Corymb
diagrammatic

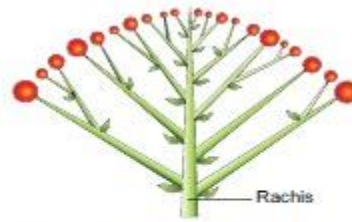


Figure 4.4: (n)
Compound corymb
diagrammatic



Figure 4.4: (o)
Corymb



Figure 4.4: (p)
Compound corymb



Figure 4.4: (q)
Umbel
diagrammatic



Figure 4.4: (r)
Compound umbel
diagrammatic

Head or Capitulum

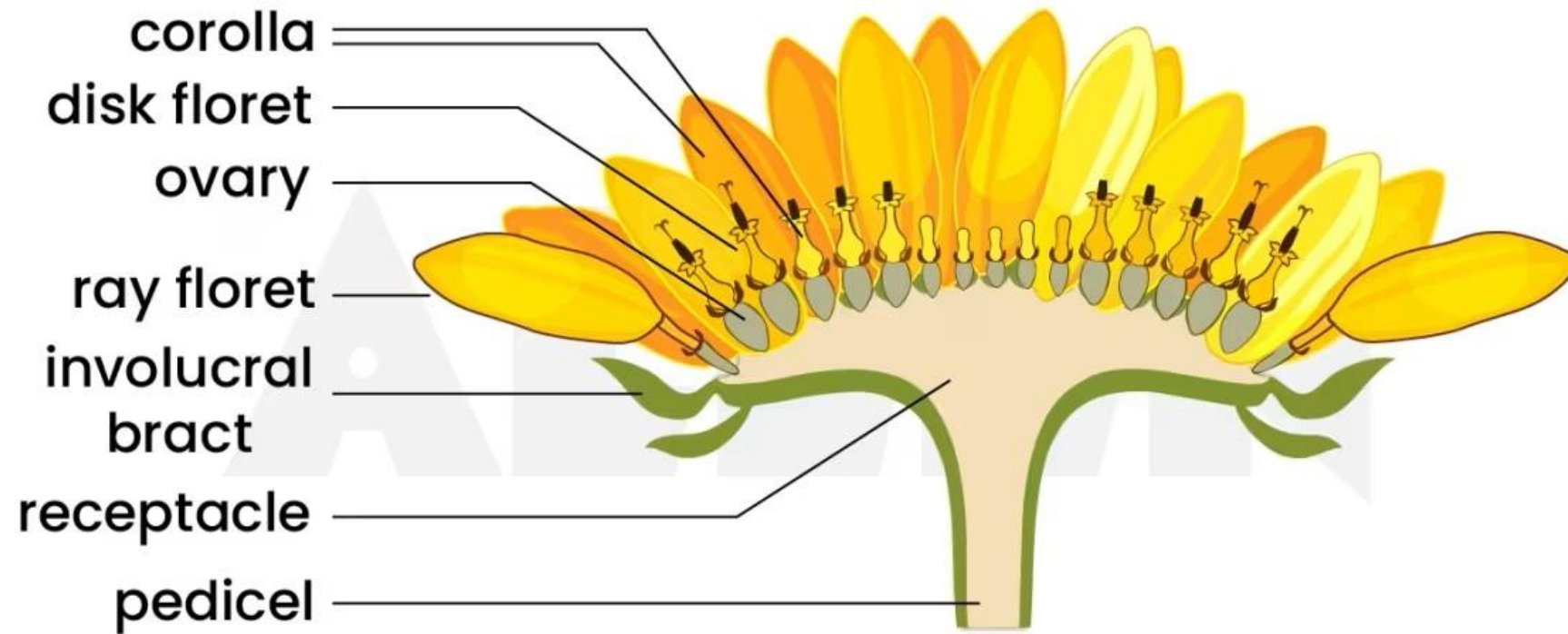
The peduncle is flattened forms a disc like receptacle from where a number of small sessile flowers called florets are developed in centripetal manner.

The whole inflorescence is covered by one or more fused bracts called as involucre.

Capitulum may be heterogamous (with ray and disc) or homogamous (all florets of one type).

Capitulum is a character of the members of family Compositae.

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- **Cymose inflorescence**

- The primary floral axis terminates into a flower and hence has definite growth. Growth continues through lateral buds. Flowers are arranged in **basipetal succession**, that is, the older flowers lie at the apex and the younger flowers below it.

- Types of cymose inflorescence are as follows:
- **Solitary cyme**
- A single, axillary or terminal flower is produced on the floral axis.
Examples: Terminal: *Datura* and Axillary: *Hibiscus*

- **Monochasial cyme**

- The main floral axis or peduncle terminates into a flower. A single lateral **branch** emerges at a time which terminates into a flower. Example: *Drosera*, *Freesia* etc.

- **Dichasial cyme**

- In dichasial cyme, the main axis terminates into a flower. Two lateral branches are produced, each of which also terminate with a flower. Example: Jasmine

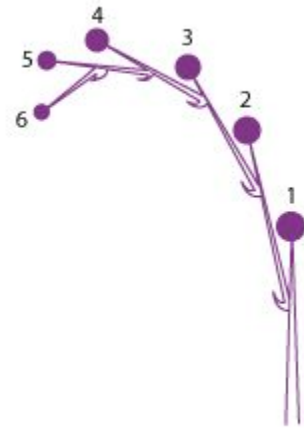
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- **Polychasial cyme**

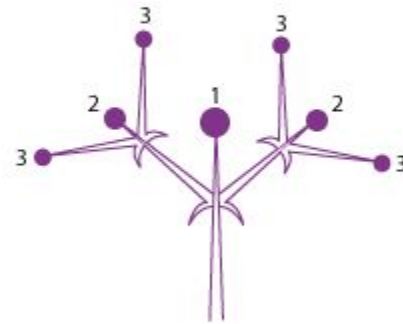
- The main/mother axis terminates into a flower but gives rise to more than two lateral branches. Each lateral branch behaves like the mother axis. Example: *Calotropis*



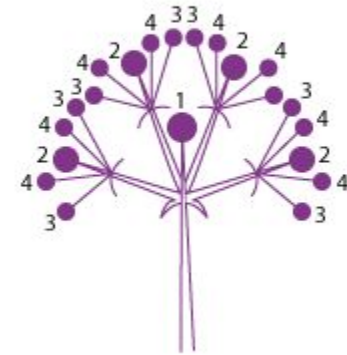
Scorpiod



Helicoid



Biparous cyme



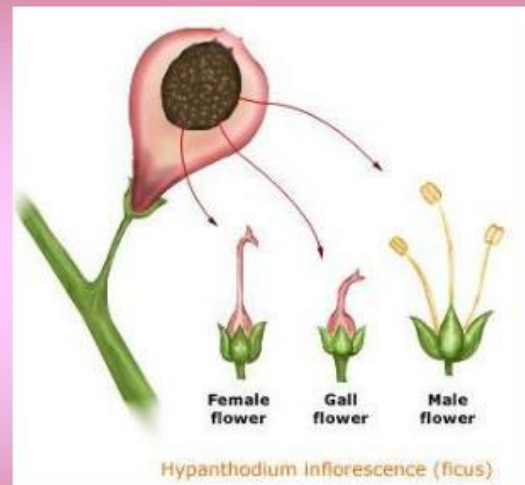
Multiparous cyme

- **Special inflorescences**

- **Hypanthodium**

- It is a special inflorescence formed by the condensation of three types of cymose inflorescence within a flask-shaped fleshy receptacle. The inflorescence has a pore lined by scales which leads to a canal. The canal has inwardly directed hair and male flowers lie towards the pore, female flowers at the base and neuter or gall flowers in between the two. This is a characteristic of the *Ficus* species.

HYPANTHODIUM



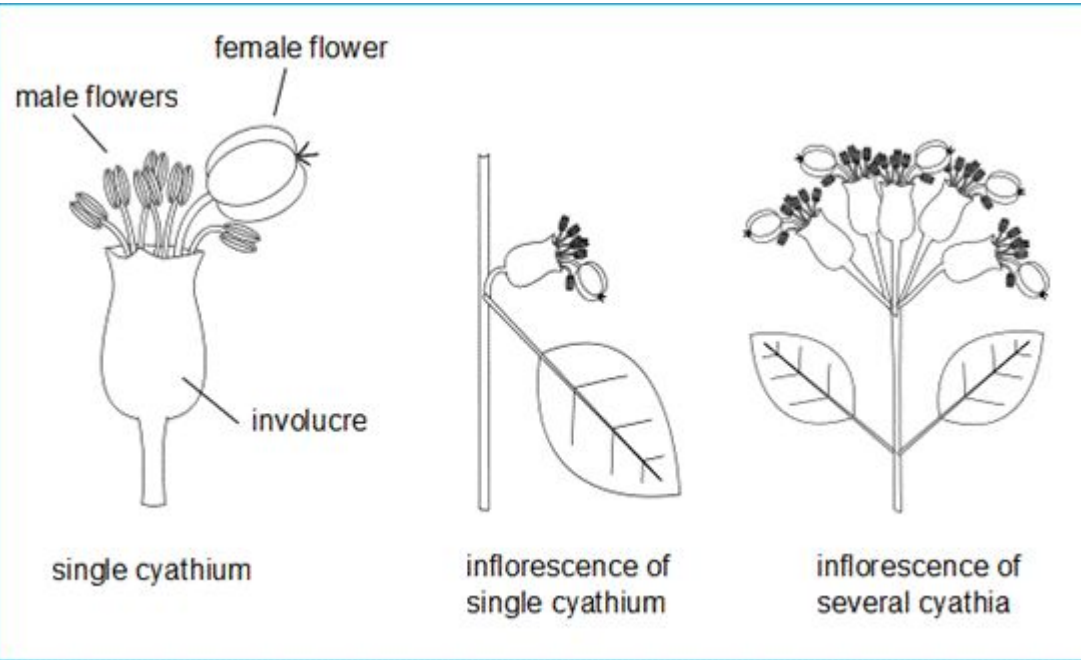
Cup-shaped receptacle formed by the condensation of the rachis of closely placed cymes. The receptacle forms a vessel with a terminal opening and flowers inside it in cymose groups e.g. *Ficus*



- **Cyathium**

- It is a type of condensed cymose inflorescence which appears like a single flower borne on a conical receptacle having green involucre of 4-5 bracts with nectariferous glands. The receptacle has a single centrally placed female flower surrounded by centrifugally placed male flowers. The flowers bear pedicels and are achlamydeous (lack both calyx and corolla) and unisexual. This indicates that flowers have a single essential organ, pistil in female flowers and stamens in male flowers. The essential organ is connected to the pedicel. This type of inflorescence is seen in *Euphorbia*, *Anthostemma*, etc.

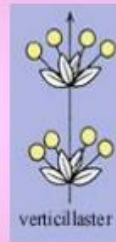
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- **Verticillaster**

- It is a raceme of whorls of cymose flowers that develop in the axils of opposite leaves. Each whorl has two clusters each of which possess 3-9 flowers formed of a biparous cyme having uniparous branching. The peduncle is reduced such that flowers appear to develop from the axil of the leaves. Due to overcrowding the two opposite clusters seem like a single whorl or verticil. This is seen in *Salvia*, *Ocimum*, etc.

VERTICILLASTER



Ocimum sanctum



The opposite leaves at the nodes, bear dichasial cymes in their axils. The sessile flowers are clustered together appearing like a whorl around the stem e.g. *Ocimum sanctum* (tulsi)

